🩺 AI101 Mini Project · Team No. 53

Diabetes Predictor

Predicting the likelihood of diabetes in patients based on age, BMI, blood glucose, HbA1c and medical history using logistic regression

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🔗 Repo: github.com/Paras-Raina/AI101-Mini-Project

🔎 Problem and data

🎯 What and why

Classify whether a patient has diabetes from age, BMI, blood glucose, HbA1c, hypertension and heart disease.

Diabetes often goes unnoticed, so a simple, explainable model can flag high-risk patients for proper testing.

🗂️ Dataset

Kaggle: Diabetes Prediction Dataset (iammustafatz)

9 columns: gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level and diabetes (target).

We use 6 as inputs and leave out gender and smoking history.

100,000
📄 raw rows

96,146
🧹 rows after removing 3,854 duplicates

8.8%
🩸 diabetic records (imbalanced)

Dataset: https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset

⚙️ Approach

📥 Data
Kaggle CSV, 100,000 rows

→

🧹 Clean
No missing values; 3,854 duplicates dropped

→

🛠️ Prepare
6 features, 80/20 stratified split, StandardScaler

→

🤖 Model
Logistic Regression (max_iter = 1000)

→

📊 Output
Diabetes yes or no, scored by accuracy, precision, recall, F1

📚 Libraries used

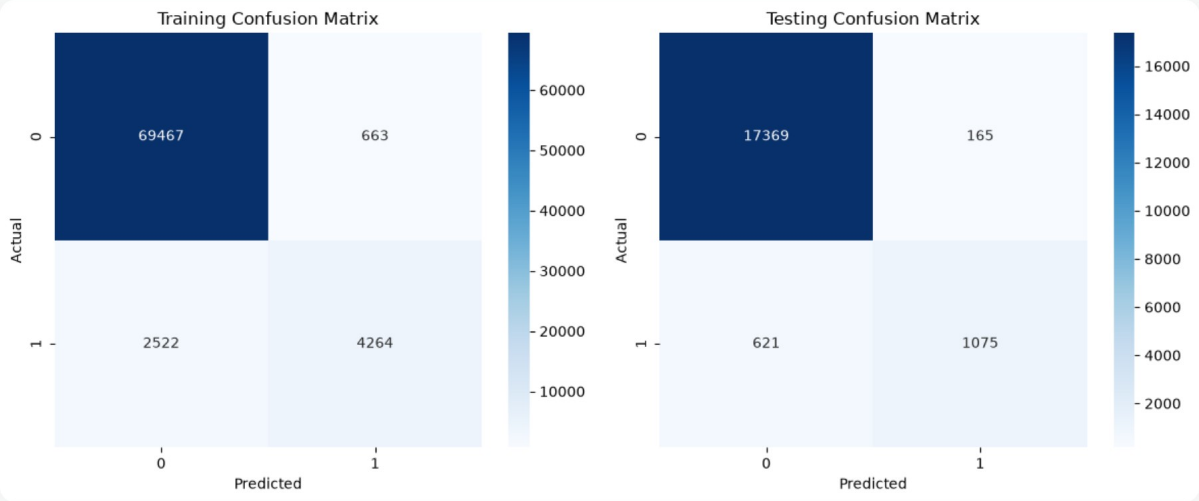
pandas · NumPy · Matplotlib · Seaborn · scikit-learn

Train set: 76,916 rows · Test set: 19,230 rows

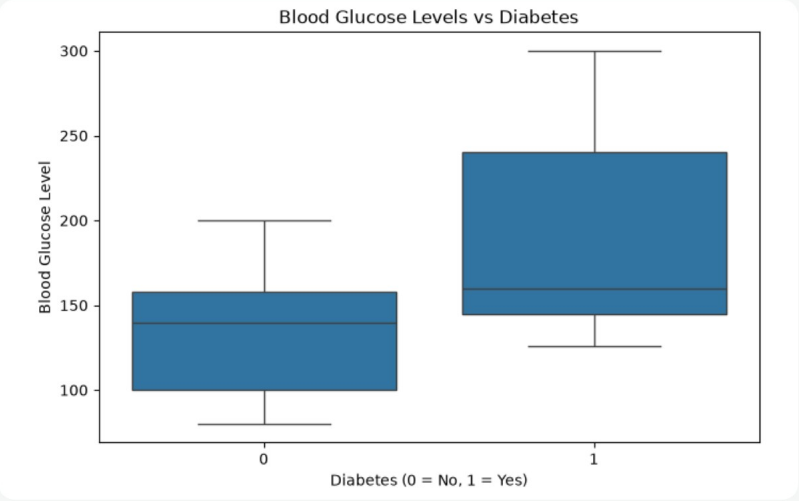
⚖️ Imbalanced data

Only 8.8% of patients are diabetic, so we used a stratified split to keep that ratio in both the train and test sets.

📈 Results



Confusion matrices: train (left), test (right)



Blood glucose is higher in diabetic patients

95.91%

✅ **Test accuracy**

Share of all patients classified correctly

86.69%

🎯 **Test precision**

When we predict diabetes, how often we are right

63.38%

🩺 **Test recall**

Share of actual diabetic patients we catch

73.23%

⚖️ **Test F1-score**

Balance between precision and recall

💡 Learnings

✅ What worked

- 95.91% test accuracy
- 86.69% test precision: most patients flagged as diabetic really were
- Train and test scores almost identical, so no overfitting
- HbA1c and blood glucose are the strongest predictors

⚠️ What didn't

- Recall is only 63.38%: 621 of 1,696 diabetic test patients were missed
- Accuracy is flattered by imbalance (8.8% diabetic)
- To try: class weights or a lower threshold

👥 Who did what

📊 **Paras Raina**
PPT Lead · Data pre-processing and cleaning

🎬 **Lakkireddy Thulasiram Reddy**
Video Lead · Model training and checking model parameters

📝 **Manav Garg**
Documentation Lead · Checked other evaluation metrics (precision, recall, F1-score)

💻 **Dhonnar Satyam Ramesh**
Execution Lead · Tested the model and added the box plot

🤖 AI-use disclosure

- **Claude:** Generated the slides the way we wanted them
- **ChatGPT:** Helped plot heat maps and write the final analysis
- **ChatGPT & Claude:** Helped generate the README and requirements.txt file